

Expand and Simplify Binomials

Pre-Uni Maths for Sciences C1.1

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P1

Part A

$$(x + 1)^4$$

Expand and simplify $(x + 1)^4$.

The following symbols may be useful: x

Part B

$$(z + 2a)^3$$

Expand and simplify $(z + 2a)^3$.

The following symbols may be useful: a, z

Part C

$$(a - b)^5$$

Expand and simplify $(a - b)^5$.

The following symbols may be useful: a, b

Find Coefficients 2

Pre-Uni Maths for Sciences C1.3

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P1

Without expanding the binomials, find:

Part A

Coefficient of x^4y^6

The coefficient of x^4y^6 in the expansion of $(x^2 + 3y^2)^5$.

Part B

Coefficient of x^{20}

The coefficient of x^{20} in the expansion of $(x^2 + 3x)^{12}$.

Part C

The coefficient of ab^7

The coefficient of ab^7 in the expansion of $(a + \frac{1}{4}b)^8$.

Constant term

The constant term in the expansion of $\left(\frac{x^2}{2} - \frac{8}{x}\right)^9$.

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STEM SMART Double Maths 27 - Binomial & Maclaurin Expansions

Group and Expand

Pre-Uni Maths for Sciences C1.5

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P1

Expand $(1 - 2x + 3x^2)^7$ in ascending powers of x as far as x^3 .

The following symbols may be useful: x

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Binomial: All Rational n 2i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Expansion

Expand $(1 - 4x)^{\frac{1}{4}}$ in ascending powers of x , up to and including the term in x^3 .

The following symbols may be useful: x

Part B

Values of a and b

The term of lowest degree in the expansion of

$$(1 + ax)(1 + bx^2)^7 - (1 - 4x)^{\frac{1}{4}}$$

in ascending powers of x is the term in x^3 . Find the values of the constants a and b .

$a =$

$b =$

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Binomial: All Rational n 1i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Partial Fractions

Given that $\frac{3x + 4}{(1 + x)(2 + x)^2} \equiv \frac{A}{1 + x} + \frac{B}{2 + x} + \frac{C}{(2 + x)^2}$, find A , B , and C .

If a value is not a whole number, enter the value as a decimal.

 $A =$ $B =$ $C =$

Part B

Expand

Hence or otherwise expand $\frac{3x + 4}{(1 + x)(2 + x)^2}$ in ascending powers of x , up to and including the term in x^2 .

The following symbols may be useful: x

Part C

Values of x

State the set of values of x for which the expansion in the above part is valid.

Construct your answer from the items below.

Items:

x

$<$

$\leq$

$>$

$\geq$

$< x <$

$\leq x \leq$

$> x \text{ or } x >$

$\geq x \text{ or } x \geq$

-4

-2

$-\frac{3}{2}$

-1

$-\frac{1}{2}$

0

$\frac{1}{2}$

1

$\frac{3}{2}$

2

4

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Maclaurin Series - Binomial

Subject & topics: Maths | Algebra | Series**Stage & difficulty:** A Level P2, Further A P1**Part A****Expand $(1 + r)^{\frac{1}{3}}$**

Expand $(1 + r)^{\frac{1}{3}}$ up to the term in r^3 .

The following symbols may be useful: r

Part B**Find $1.1^{\frac{1}{3}}$**

Hence, using your expansion, find $(1.1)^{\frac{1}{3}}$ to 3 decimal places.

Part C**Find $9^{\frac{1}{3}}$**

Now, using your expansion again, find $9^{\frac{1}{3}}$ to 2 decimal places.

Part D

Electric field on the axis of a charged sheet

The electric field E on the axis of a uniformly charged circular sheet at a distance z from the centre of the sheet is given by

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{z^2 + a^2}} \right]$$

where σ is the charge per unit area on the sheet and a is the radius of the sheet. Show that in the limit when $z \gg a$ the field on the axis is such that $E \approx \frac{A}{z^2}$ and find A .

The following symbols may be useful: A, a, epsilon_0, sigma, z

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STEM SMART Double Maths 27 - Binomial & Maclaurin Expansions

Maclaurin Series - Cos & Sin 1

Pre-Uni Maths for Sciences C3.6

Subject & topics: Maths | Algebra | Series

Stage & difficulty: Further A C2, University C1

Part A

Find the cosine of the angle 0.2 rad

Find, using a Maclaurin expansion, the cosine of the angle 0.2 rad, correct to 3 decimal places.

Part B

Find the sine of the angle 0.08 rad

Find, using a Maclaurin expansion, the sine of the angle 0.08 rad, correct to 2 significant figures.

Part C

Potential energy of mass on pendulum

A pendulum consists of a point mass m suspended on a light string of length l . When the string makes an angle of ϕ to the vertical its potential energy relative to the point where $\phi = 0$ is given by $mgl(1 - \cos \phi)$. Show that for $\phi \ll 1$ the potential energy is given approximately by $A_0\phi^2$ and find an expression for A_0 .

The following symbols may be useful: g , l , m

Maclaurin Series - ln

Pre-Uni Maths for Sciences C3.2

Subject & topics: Maths | Algebra | Series

Stage & difficulty: Further A P2, University P1

Part A

Expand $\ln(1 + z)$ and hence $\ln(2 + 4y)$

(i) Write down the Maclaurin expansion of $\ln(1 + z)$ up to the term in z^3 .

The following symbols may be useful: z

(ii) By re-writing $\ln(2 + 4y)$ in the form $A + \ln(1 + z)$, where A is a constant, find the Maclaurin expansion of $\ln(2 + 4y)$ up to the term in y^3 .

The following symbols may be useful: y , z

Part B

Expand $\ln\left(\frac{1+q}{1-q}\right)$

Find the first 4 non-zero terms in the Maclaurin expansion of $\ln\left(\frac{1+q}{1-q}\right)$.

The following symbols may be useful: q

Maclaurin Series - Exponentials 3

Pre-Uni Maths for Sciences C3.5

Subject & topics: Maths | Algebra | Series

Stage & difficulty: Further A C2, University C1

Part A

Expand $Ae^{-\alpha t}$

Expand $Ae^{-\alpha t}$ up to the term in t^2 .

The following symbols may be useful: A, alpha, p, t

Part B

Expand $e^p - e^{-p}$

Find the first two non-zero terms in the Maclaurin expansion of $e^p - e^{-p}$.

The following symbols may be useful: A, alpha, p, t

Part C

Energy decay in oscillations

A lightly damped oscillatory system has a period T . The total energy of the system at time t is given by $E(t)$. One period later its energy $E(t + T) = E(t)e^{-\gamma T}$.

(i) Find an expression for the fractional change in energy in one cycle.

The following symbols may be useful: T , e , γ

(ii) On the assumption that $\gamma T \ll 1$ find an approximate expression for the fractional change in energy in one cycle.

The following symbols may be useful: T , e , γ

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